From helpful answers to lasting understanding.

A learning platform built around what students can do independently.

COFOUNDER ONBOARDING

Engine foundation today.
A student pilot next.



Understanding an explanation is only the start.

01

I can follow it.

Useful help in the moment

→

02

I can do it.

A fresh problem, independently

→

03

I still know it.

Evidence after a delay

Our bet: connect all three into one focused learning experience.

Start narrow. Build for a broader horizon.

The first commercial segment is still a decision.

LONG-TERM VISION

Primary → high school → early university

PILOT HYPOTHESIS

One student group. One curriculum. One need.

BUILT FOUNDATION

Maths / Quadratics

12 skills • Practice engine

More subjects need their own content + checks.

One useful session, connected to the next.

1

Try

A focused problem

2

Ask

Your work + your question

3

Get help

One useful way forward

4

Show it

Try independently

5

Return

A fresh check after a delay

Engine support exists • Student interface still to build

PLANNED

Four modes. One implemented engine path.

Practice

ENGINE BUILT

Attempt. Ask. Adapt.

Review

FOUNDATIONS ONLY

Retrieve before forgetting.

Learn

PLANNED

Find your starting point. Build from there.

Test

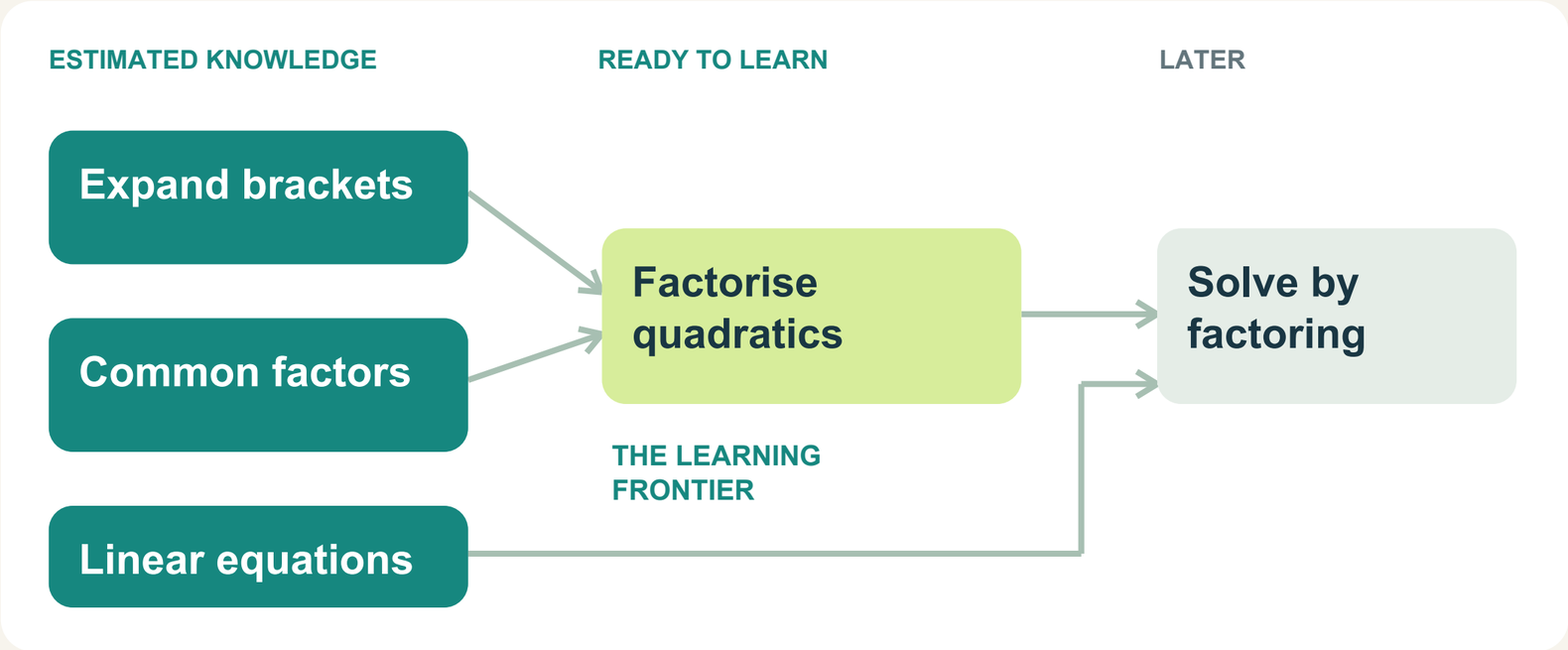
PLANNED

Demonstrate independence.

Learn: start at the edge of what you know.

“What am I ready to learn next?”

PLANNED



Locate → build → check

01 Gauge your starting point

Use evidence + targeted checks.

02 Teach the next ready skill

Worked examples → less help.

03 Update the picture

Try independently; check later.

Illustrative graph • Readiness can differ by branch; weak prerequisites need repair.

Practice: turn understanding into capability.

“Can I use what I have learned?”

ENGINE BUILT

ILLUSTRATIVE EXERCISE

$$x^2 - 5x + 6 = 0$$

“I have (x − 2)(x − 3).
What do I do next?”

Working + a focused question

HELP THAT LEAVES A NEXT STEP

“What does a product of zero
tell you about the factors?”

Try again → build fluency → vary the problem

Record how much help the attempt needed.

Help is welcome. Assisted success and independent evidence stay distinct.

Review: keep previous learning available.

“Can I still recall and use this?”

MODE PLANNED

01

Learned earlier

Previous independent evidence

02

Due for retrieval

A card or a fresh problem

03

Update the schedule

Use recall evidence to plan ahead

Forgot something?

Brief correction + reschedule.
Choose Practice if you want to work on it now.

Test: see what you can do without help.

“Where am I independent, and where are the gaps?”

PLANNED

DURING THE ASSESSMENT

Skill A | **Skill B** | **Skill C**

Attempt independently • No hints or tutor feedback

Mock exam: add timing + an exam blueprint.

→

AFTER FINISHING

Your diagnostic report

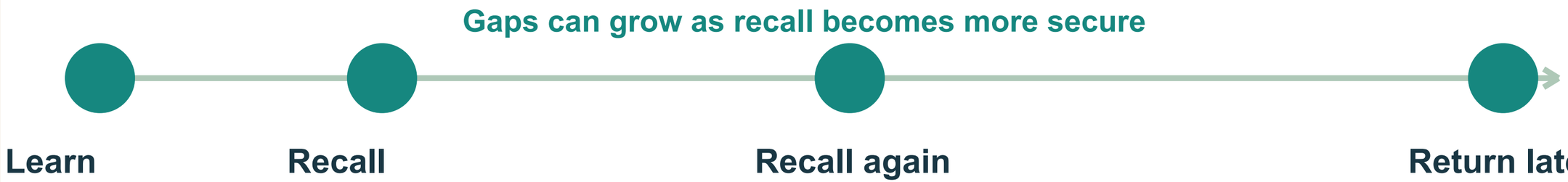
Strengths • Gaps
What to work on next

Assess first. Explain the results and plan the next learning steps afterwards.

Revisit. Retrieve. Reinforce.

Spaced repetition + retrieval practice: research-backed principles.

ILLUSTRATIVE SUCCESSFUL-RETRIEVAL PATH



A lapse brings the next check closer. Timing adapts; this is not a fixed calendar.

SPACING

Revisit across time.

RETRIEVAL

Try recalling before seeing the answer.

Memory foundations built • Review scheduler planned • LearnAI outcomes still to validate

Research: Cepeda et al. (2008); Roediger & Karpicke (2006). Links and scope in presenter notes.

Elo: match the challenge to the learner.

A rating method from chess, adapted to learners and exercises.

CORE BUILT

Compare

Skill strength
with exercise difficulty

→

Predict

Estimate the chance
of an unaided success

→

Update

Compare the result
with that prediction

Surprising success → larger increase. Expected success → smaller increase.

Skill strength → how challenging?

Memory freshness → when to revisit?

Per-skill estimate • Assistance-aware evidence • Calibration still needs student data

Educational precedent: Klinkenberg, Straatemeier & van der Maas (2011). Source and limits in notes.

A substantial foundation already exists.

Implemented in the development workspace; not a deployed service.

12
quadratics skills

12
exercise templates

2
AI providers

Learning engine

Prerequisites
Adaptive practice
Assistance-aware evidence

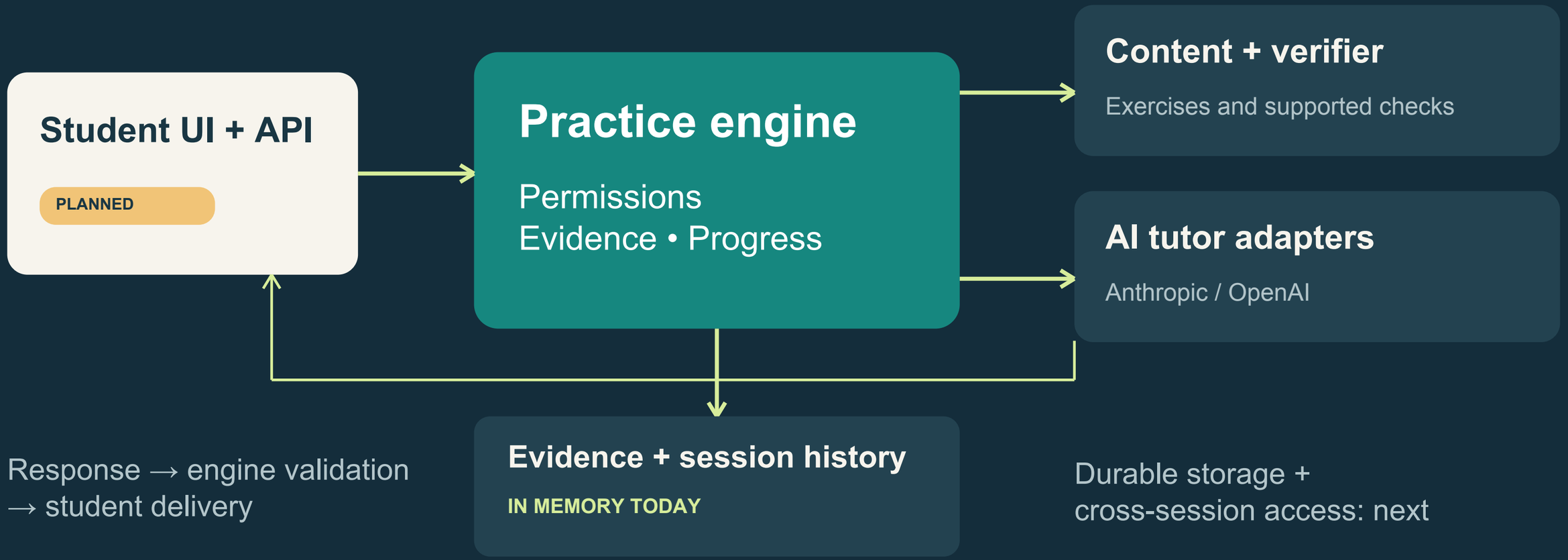
Grounded interaction

Maths + partial-work checks
Revision-aware context
Controlled help

Quality tooling

Bounded runtime
Saved traces + replay
Regression + live evaluation

The engine decides. The model explains.



Response → engine validation → student delivery

Durable storage + cross-session access: next

Encouraging signals. A small evidence base.

1,869

keyless tests passed

Latest recorded full suite
4 live probes deselected

13 / 14

live replies accepted

One delivery failure
Small scenario batch

2.7s

median accepted latency

13 accepted replies
Not a production guarantee

Software evidence ≠ learning outcomes. Automated judge comparison is still pending.

An accepted answer can still be wrong.

REVIEWED CASE · PARAPHRASED

$$x - 2 = 0$$

Student asks: “Subtract 2?”

Tutor endorsed it.
The useful step was to add 2.

Three priorities

01 Clarify ambiguity

02 Check the actual advice

03 Leave useful work to do

Delivery checks are a guardrail, not a guarantee of teaching quality.

Make explanations readable by design.

A shared content contract for the engine and the front end.

DESIGN DRAFT

PARAGRAPH

One clear idea.

LIST

A short sequence of steps.

EQUATION

$x^2 - 5x + 6 = 0$



PRODUCT CONCEPT

Look for two numbers.

- Their product is 6.
- Their sum is −5.

Which pair would you try?

Text, lists and maths first. Interactive elements later.

Turn the engine into one useful student session.

01 A usable interface

Maths input • Work • Feedback

02 A persistent service

API • Auth • Database • History

03 Operational confidence

Recovery • Monitoring • Cost

04 A bounded pilot

Learners • Content • Data care

Two workstreams, one path to a pilot.

Dependency order, not delivery dates.

ENGINEERING

Quality baseline

Judge comparison
Failure follow-up

→

Presentation

Readable content
Validated rendering

→

Practice product

UI + API + storage
Resumable sessions

→

Student pilot

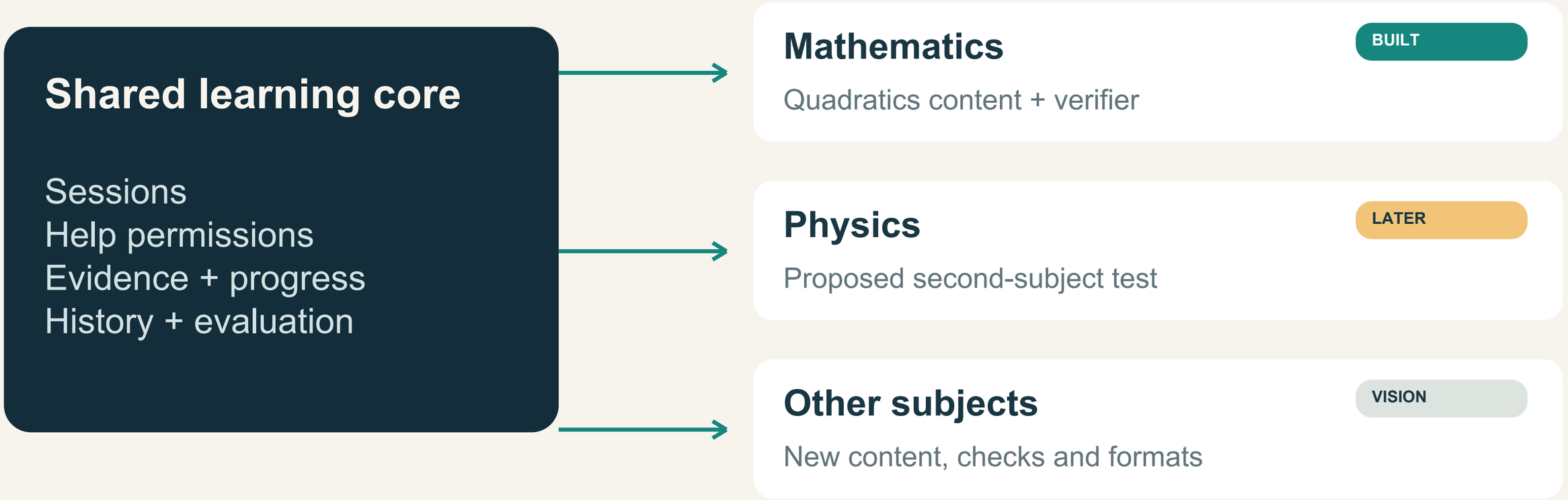
Observe real use
Measure outcomes

PRODUCT + BUSINESS — START NOW

Choose the learner → Prototype input → Validate buyer → Recruit + learn

After the pilot: Review → Learn → curriculum & assessment → another subject

Keep the core generic. Prove each subject.



Extensibility is a design direction. A second subject must validate it.

Measure what students can do later.

Primary hypothesis: stronger independent performance after a delay.

01 Learning

Fresh unaided success
Delayed success

02 Experience

Session completion
Maths-input friction

03 Reliability

Wrong or missing feedback
Recovery + latency

04 Business

Repeat need • Willingness to pay
Cost per useful session

Choose one learner.
One problem.
One useful session.

WHO?

Student • Curriculum
Buyer • Distribution

WHAT?

Learning moment
Input experience

WHAT PROVES IT?

Pilot evidence
Value + viable cost

Our next job: turn a strong foundation into a testable product promise.